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## 1 SUPERCHARGING INDIA'S EV POWER PLAY

EVs are booming, but battery reliability remains a bottleneck. One startup is rethinking how batteries are built to improve strength, consistency, and performance.



IBC cell and pack

As electric vehicles gathered momentum and the transition to clean energy accelerated, Priyadarshi Panda identified a critical inflection point. He recognised that battery manufacturing mirrors semiconductor production in its core challenges: process precision, consistency, and long-term reliability.

In 2022, Priyadarshi started International Battery Company

(IBC) based in the heart of Silicon Valley in the US, which is the global innovation engine, with operations in South Korea and India. "In India, we operate through a majority ownership joint venture with Mahanagar Gas Limited, called IBC India Private Limited," says Priyadarshi.

The company specialises in three main areas of lithium-ion battery technology. First, they focus on material innovation, developing

new battery materials to improve performance and reliability. Second is battery design and device integration, which covers chemistry, electrochemistry, and the internal configurations, like jelly rolls, to meet the needs of applications ranging from telecom and solar grids to high-performance defence, AI data centres, and vehicles of all sizes. Third is product assembly, combining these devices or device-ready components into finished batteries, mostly in prismatic form, which accounts for 70-75% of lithium-ion deployments today and is expected to grow to > 80% by 2028.

On talking about the innovation, Priyadarshi explains, "We are molecule innovators and largely chemistry-agnostic product-minded technologists who bring significant scaled manufacturing experience to be able to incorporate design for high-speed manufacturing into all aspects of the development. In mobility applications, we work with high-power, high-energy cathodes like nickel-cobalt-manganese (NMC) and nickel-cobalt-aluminium (NCA) oxides, which are water-sensitive and require controlled dry-room environments. On the anode side, we have developed electrodes which use non-Chinese graphite and also variants which blend graphite with higher-specific-capacity silicon-based materials to optimise performance. For stationary storage, energy density is less important than cycle life, since grid projects often last 20+ years. Here, we use lithium iron phosphate (LFP) with larger particle sizes on the cathode and cost-effective graphite on the anode."

IBC's core product line is built around high-performance prismatic lithium-ion cells. These cells come in various chemistries, including NMC,